



Slides for P3568R1

break label; and **continue label;**

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Source:

github.com/Eisenwave/cpp-proposals/blob/master/src/break-continue-label-slides.cow



IMPORTANT: This document has custom controls:

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break label;
and
continue label;
P3568R1

Introduction

C2y now supports `break` and `continue` with a label (N3355):

```
outer: for (int i = 0; i < N; ++i)
    for (int j = 0; j < M; ++j)
        if (/* ... */)
            break outer; // OK, break targets "outer"
```

- would be useful in C++, not just in C
- if C has it, EWG wants it
- clearly implementable, even in `constexpr`

Recent history

Hagenberg 2025

EWG likes syntax N3355: `for (...) { }`

SF	F	N	A	SA
4	16	5	9	5

EWG likes syntax `for` N3377 `(...) { }`

SF	F	N	A	SA
7	13	5	5	8

If C has it, we are interested in this feature too.

SF	F	N	A	SA
16	21	5	2	3

Conclusion

Let WG14 figure out syntax;
we'll copy the feature into C++.

Recent history

Graz 2025

Would WG14 like to see a paper changing loop name syntax at a future meeting?

F	N	A
6	11	9

- **Conclusion:** N3355: `for (...) { }` syntax wins
 - however, *major* N3377 concerns unaddressed
- C++ goes with N3355 syntax
 - also let's try to fix N3377 issues before WG14

Unaddressed N3377 concerns

Labels must be unique per function $\Rightarrow$ problematic for macros:

```
#define MACRO() \  
    outer: for (/* ... */) \  
        for (/* ...*/) break outer  
// ...  
MACRO(); // OK  
MACRO(); // error: duplicate "outer"
```

Also, nested reuse of names impossible:

```
outer: for (/* ... */) inner: for (/* ... *)  
    outer: for (/* ... */) inner: for (/* ... */) // error
```

P3568R1 strategy

- labels can go *anywhere*, even multiple times
- restrict `goto`, `break`, `continue`, not labels

```
label: label: while (true) break label; // OK
label: for (/* ... */) continue label; // OK

goto label; // error: jump to duplicate label
while (true) break label; // error: break of unrelated loop
```

- **Idea:** labels are *basically* comments, so why restrict them?
- **Benefit:** very simple, teachable system

Summary

- EWG wants `break label`, and we have a design
- syntax debate is resolved
- N3377 concerns can be addressed with `label : syntax`
- wording exists, CWG chair looked at it
- implementation non-trivial, but *clearly* possible
 - GCC patch for N3355 exists (C fronted)
 - Clang implementers convinced that `break label` is feasible